

Abstract

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers?8x deeper than VGG nets but still having lower complexity. An ensemble of these residual nets achieves 3.57% error on the ImageNet test set. This result won the 1st place on the ILSVRC 2015 classification task.

1. Introduction

Deep convolutional neural networks have led to a series of breakthroughs for image classification. When deeper networks are able to start converging, a degradation problem has been exposed: with the network depth increasing, accuracy gets saturated and then degrades rapidly. Unexpectedly, such degradation is not caused by overfitting, and adding more layers to a suitably deep model leads to higher training error.

2. Related Work

Residual Representations. In image recognition, VLAD is a representation that encodes by the residual vectors with respect to a dictionary, and Fisher Vector can be formulated as a probabilistic version of VLAD. Both are powerful shallow representations for image retrieval and classification.

Shortcut Connections. Practices and theories that lead to shortcut connections have been studied for a long time. An early practice of training multi-layer perceptrons (MLPs) is to add a linear layer connected from the network input to the output.

3. Methodology

Residual Learning. Let us consider $H(x)$ as an underlying mapping to be fit by a few stacked layers (not necessarily the entire network), with x denoting the inputs to the first of these layers. Rather than expecting stacked layers to approximate $H(x)$, we explicitly let these layers approximate a residual mapping $F(x) := H(x) - x$. The original mapping is recast into $F(x) + x$. We hypothesize that it is easier to optimize the residual mapping than to optimize the original, unreferenced mapping.

Identity Mapping by Shortcuts. We adopt residual learning to every few stacked layers. A building block is defined as: $y = F(x, \{W_i\}) + x$. Here x and y are the input and output vectors of the layers considered. The function $F(x, \{W_i\})$ represents the residual mapping to be learned. The operation $F + x$ is performed by a shortcut connection and element-wise addition.

4. Datasets and Experiments

ImageNet Classification. We evaluate our method on the ImageNet 2012 classification dataset that consists of 1000 classes. The models are trained on the 1.28 million training images, and evaluated on the 50k validation images. We also obtain a final result on the 100k test images. We evaluate both top-1 and top-5 error rates.

Baseline Networks. Our baseline comparison utilizes plain networks inspired by VGG network philosophy (3x3 conv filters, downsampling by conv layers with stride 2).

ResNet Architecture. Based on the plain networks, we insert shortcut connections which turn the network into its residual counterpart. The identity shortcuts can be directly used when the input and output are of the same dimensions.

5. Results and Quantitative Findings

Our 18-layer and 34-layer plain nets show degradation: the 34-layer plain net has higher training and validation error (28.54% top-1 error) than the 18-layer plain net (27.94%).

In contrast, for ResNets: the 34-layer ResNet achieves a top-1 error rate of 25.03%, outperforming the 18-layer ResNet (27.88%) by 2.85%.

Deeper ResNets (50-layer, 101-layer, and 152-layer) achieve significantly lower error rates. The 152-layer ResNet achieves top-1 validation error of 19.38% and top-5 error of 4.49%.

On the ImageNet test set, an ensemble of ResNets achieves a top-5 error of 3.57%, winning 1st place in ILSVRC 2015.

6. Discussion and Limitations

While residual learning enables training of ultra-deep networks up to 1000+ layers on CIFAR-10, we observe that extremely deep networks (e.g. 1202 layers) suffer from mild overfitting due to high capacity on small training sets. Furthermore, shortcut connections introduce memory overhead during backward backpropagation passes on commodity GPU clusters.

7. Conclusion and Future Work

We have presented a residual learning framework for image recognition. Our deep residual networks show strong generalization on ImageNet and COCO object detection and segmentation tasks. Future directions include exploring non-identity projection shortcuts, neural architecture search for residual blocks, and applying residual connections to natural language processing sequence models.

References

- [1] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. ICLR, 2015.
- [2] Y. LeCun, B. Boser, et al. Backpropagation applied to handwritten zip code recognition. Neural computation, 1989.
- [3] A. Krizhevsky, I. Sutskever, and G. E. Hinton. ImageNet classification with deep convolutional neural networks. NIPS, 2012.
- [4] C. Szegedy, W. Liu, et al. Going deeper with convolutions. CVPR, 2015.